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1
2 function [echogram70,fEcho,RangeBinned,pulse_length] =✓
calcEchogram70kHz(Config, Data, WaterProp, Options)
3 % function [echogram70] = calcEchogram70kHz(Config, Data)
4
5 %% INIT
6 echogram70 = [];
7
8 try
9     soundVel = WaterProp.soundVel;
10 catch
11     soundVel = 1500;
12 end
13
14 soundVel2 = soundVel/2;
15
16 if isempty(Options)
17 else
18 end
19
20
21 %% CALC
22 fAdjMonochromeAmp = 6.0206; % Adjustment used in firmware [dB]
23
24 try
25     BD=Config.echo_blanking; %Reference to non-existent field✓
    'echo_blanking'.
26 catch
27     BD=Config.EchoSounder_BlankingDistance;
28 end
29
30
31 if (Config.echo_frequency1 == 70)
32     nEcho=1;
33 else
34     if isfield(Config,'echo_frequency2') && Config.echo_frequency2 ==✓
70
35         nEcho=2;
36     else
37         if isfield(Config,'echo_frequency3') && Config.echo_frequency3✓
== 70
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38         nEcho=3;
39     else
40         disp('No 70 kHz data');
41         return;
42     end
43 end
44 end
45
46 eval(['sampleRate = Config.echo_rawSampleRate' num2str(nEcho) ';''])
47 eval(['nRawSamp    = Config.echo_nRawSamp'      num2str(nEcho) ';''])
48 eval(['DataI = Data.RawEcho' num2str(nEcho) ' _70kHz_DataI;'])
49 eval(['DataQ = Data.RawEcho' num2str(nEcho) ' _70kHz_DataQ;'])
50 eval(['Range = Data.RawEcho' num2str(nEcho) ' _70kHz_Range;'])
51 eval(['fEcho = Data.Echo' num2str(nEcho) ' 70_0kHz_Frequency(1);'])
52 eval(['pulse_length = Config.echo_transmitLength' num2str(nEcho) ✓
';'])
53
54 if nRawSamp > 0,
55     sampDist70kHz = soundVel/(2*sampleRate);
56     Nsamp = size(DataI,1);
57
58     % DSP processing
59     startInd70 = 1+round((BD-Range(1))/sampDist70kHz);
60 else
61     disp('No raw echo data');
62     return;
63 end
64
65 sampDist70 = soundVel2/sampleRate;
66 samplesPerCell = Config.echo_binSize/sampDist70;
67 BinLength=zeros(1,Config.echo_numBins);
68
69 % Calculate the number of samples per bin
70 BinLength(1) = round(samplesPerCell);
71 for i=2:Config.echo_numBins
72     BinLength(i) = round((i*Config.echo_binSize - sum(BinLength(1:i-✓
1))*sampDist70)/sampDist70);
73 end
74
75 echogram70 = zeros(Nsamp, Config.echo_numBins);
76 RangeBinned = zeros(1, Config.echo_numBins);
```

```
77 for n=1:Nsamp
78     k = startInd70; % Sample number for start of first echogram bin.
79
80     for m=1:Config.echo_numBins
81         samplesInCell = k:(k+BinLength(m)-1);
82
83         Z = DataI(n,samplesInCell)+j*DataQ(n,samplesInCell);
84         sumZZ = sum(Z.*conj(Z));
85         if sumZZ == 0,
86             % Get rid of log10 of zero messages
87             sumZZ = 1e-16;
88         end
89         echogram70(n,m) = fAdjMonochromeAmp + 10*log10(sumZZ/BinLength
(m)); % Average. Using 10*log10 since Z is squared ✓
90
91         RangeBinned(m) = mean(Range(samplesInCell));
92
93         k = k+BinLength(m);
94     end
95 end
96
97 zz=0;
98
```